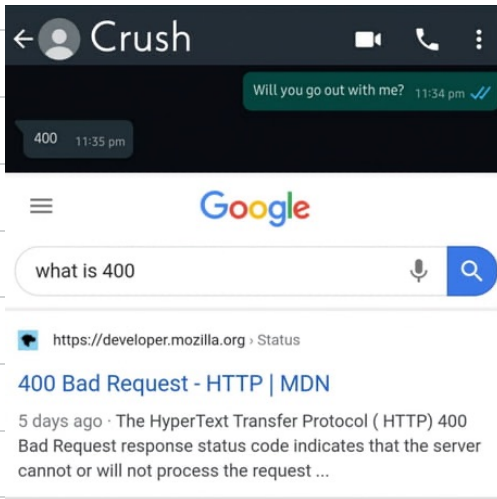


DEVELOP WEB APPS USING FLASK

Jan 08,
2024

Session - 3



AGENDA

- ① How does WWW works?
- ② What is API
- ③ What is Flask, why do we need it??

www



www.google.com

address

IP address

142.251.32.36

DNS

★ HTTP - Status codes

HTTP status codes in a nutshell:

Or, what a 404 error & all of its friends mean in simple human words.

1xx Hold on.

2xx Here you go.

3xx Go away.

4xx You screwed up.

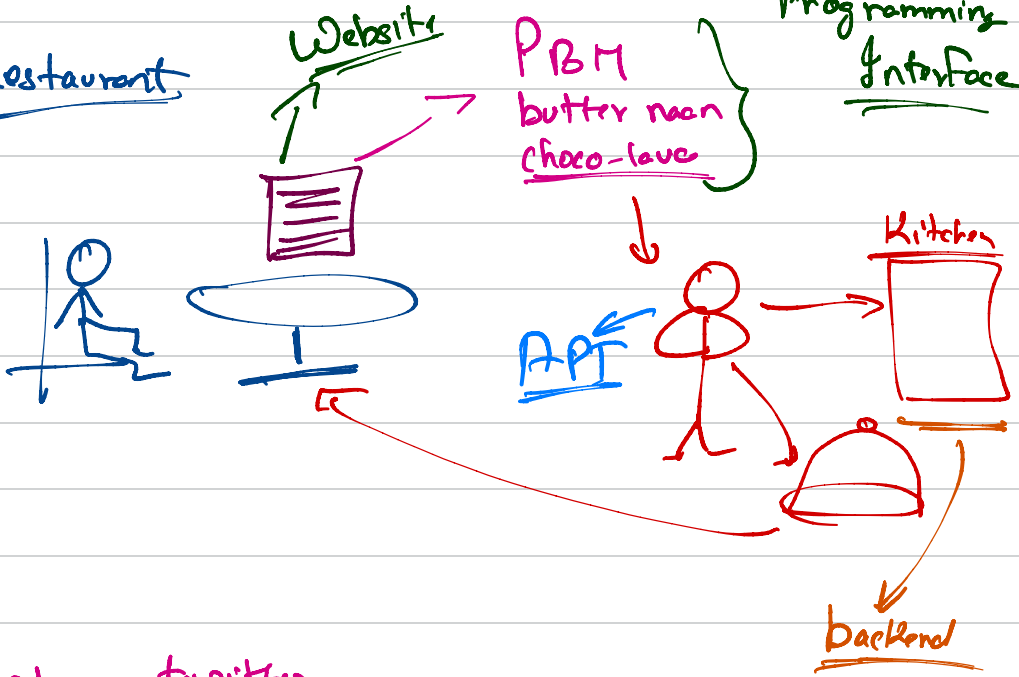
5xx I screwed up.

} g → Software engineer who made the website.

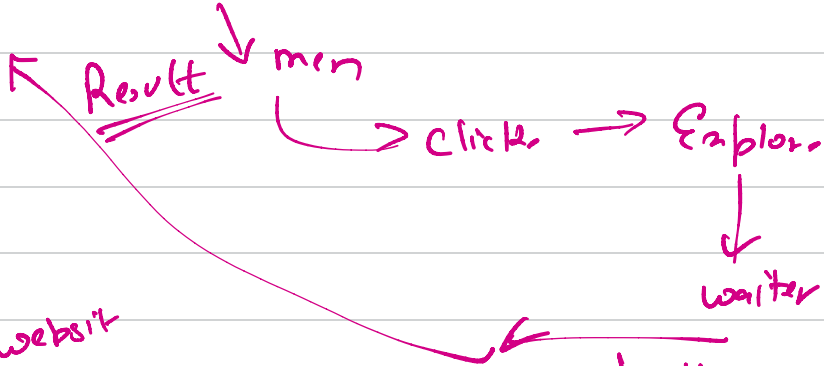


API → Application Programming Interface

Restaurant



goto twitter



website

backdoor

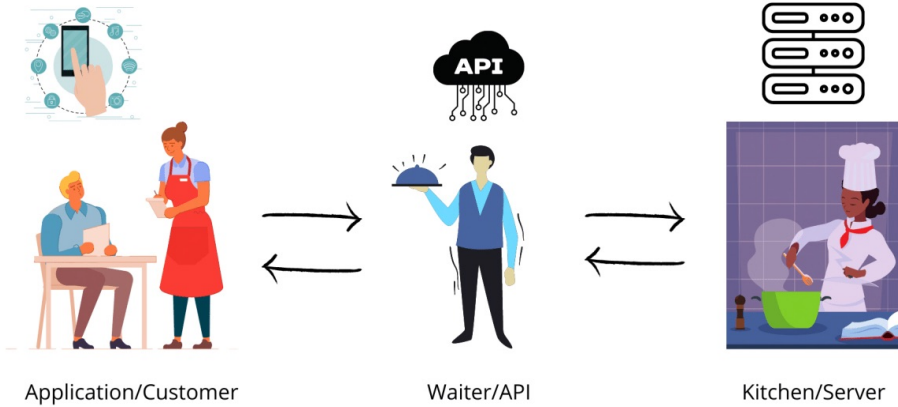
backdoor

API

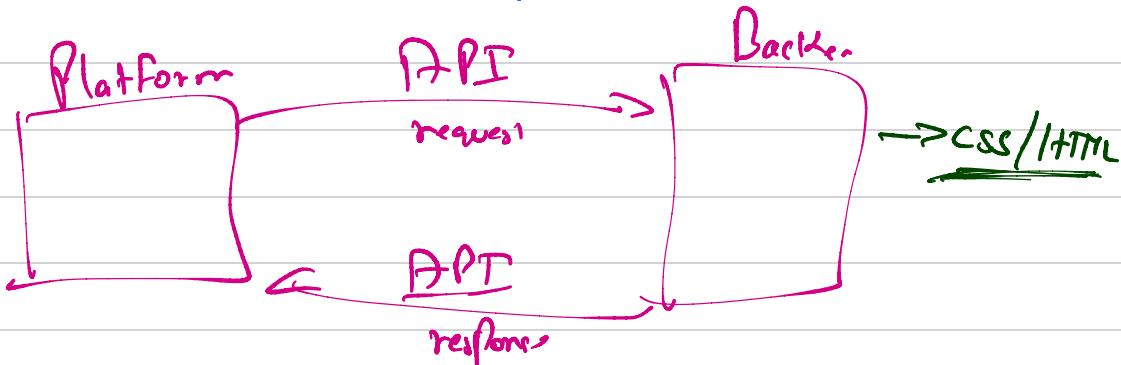
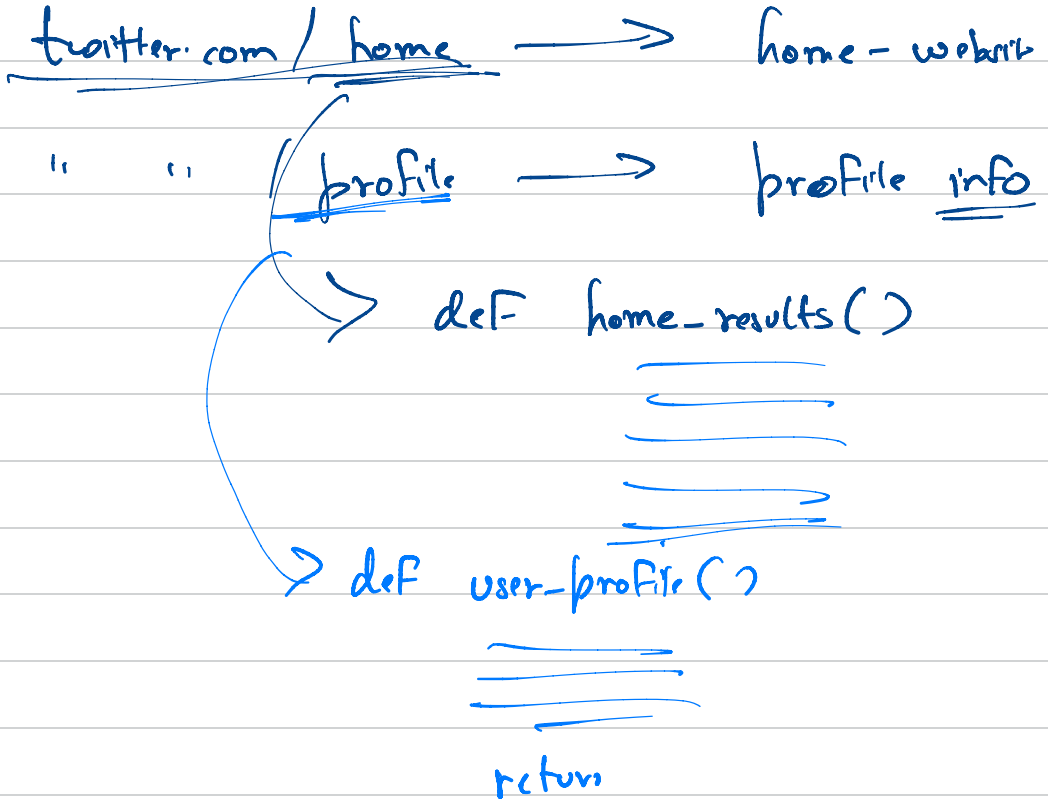
Client



What is API ?



Full Example → Twitter



set of rules → maps

twitter.com/home → funcn



① 1+HTTPS

* Set of protocol for how this communication should happen

* Makes sure → encrypted

② GET/POST

→ GET → Fetch some information from Internet

POST → Send some information to Internet

→ Check Instagram Feed → GET
→ " " profile → GET
→ Like " post → POST

→ Get requests are tracked in history
Post " " not " " "

click → like button → API - trigger
↓
Internal function
↓
adds +1 to user like

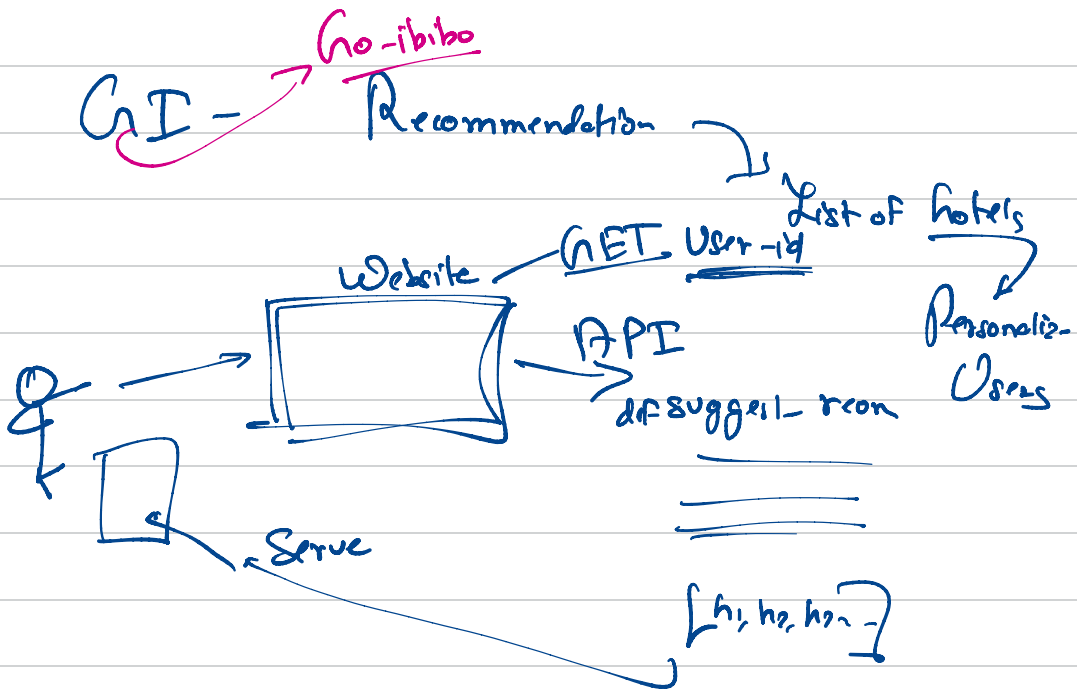
In

GET → all the param. sent by you
shows up in URL

POST → Param → are added in body
of the URL

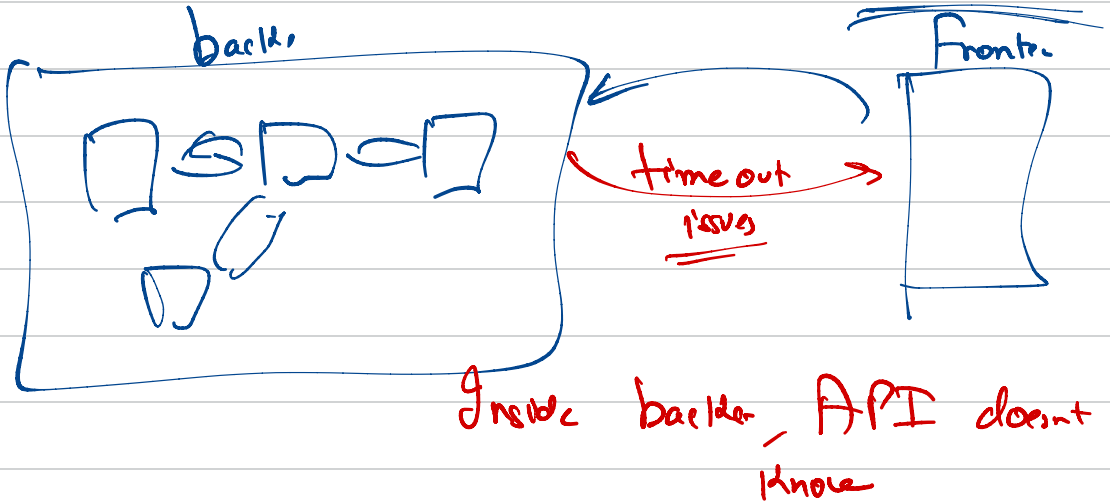
WEB SCRAPING.

- ① Path
 - ② Selenium
 - ③ requests
 - ④ Scrapy
 - ⑤ BeautifulSoup
- Prime pillar
- optional



AP $\rightarrow$ triggers function

Server
operation



When you hit API $\downarrow$

threshold $\rightarrow$

Not Fr
 $\downarrow$
50ms

$\downarrow$
func to
revert

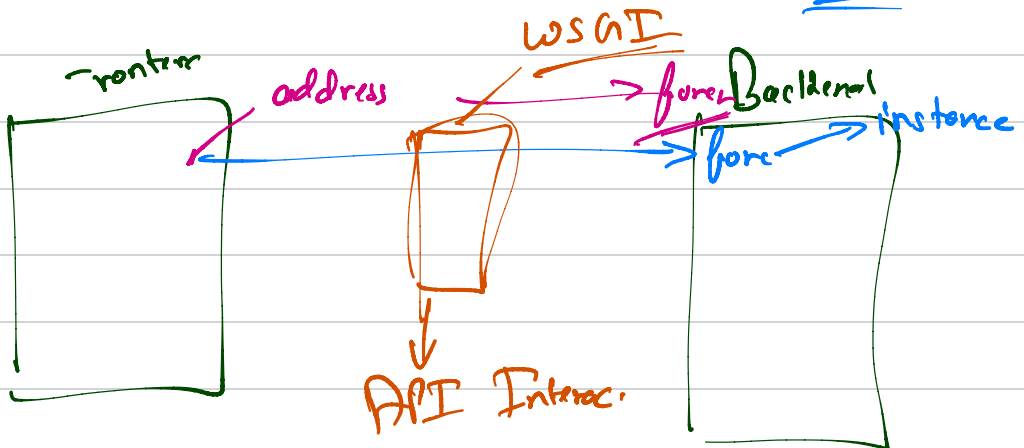
> 50 sec $\rightarrow$ gateway
timeout error

500

① Load balancing.

4 crore → 10000
online always??
200 servers

Load balancer → based on traffic
inc or dec no of
servers

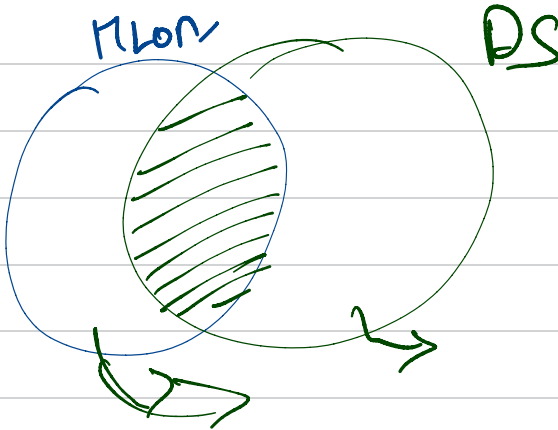


Load balancer gone

M2-System
Perf - 1/2

Body $\rightarrow$ separate from URL

{ "user-id", xyz,
"post-id", abc }



15 letters/man
50K /month

★ FLASK

→ Streamlit → only for POC

→ not scalable → 1/10th Usage

→ Light weight web application Framework

↳ Just backend.

↳ Need to build Frontend separately.

→ Django → heavy weight WAF
web application Framework

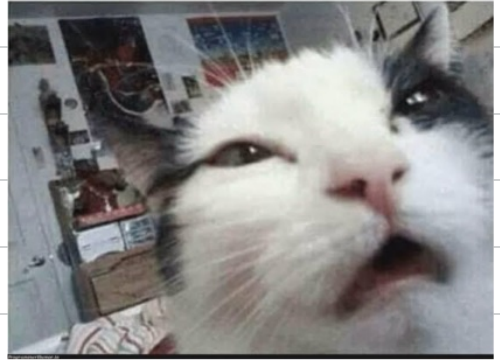
→ HDFC Bank

→ deploy ml model → Loan approval
project

twitter.com / home
base-uri additional - end-point

twitter is very volatile

Me in the morning checking if
my Twitter API app is still working.



ff

}}

models $\rightarrow$ predict

batch-size $\times$ 10

1 data sample $\rightarrow$ 1

$$b \times \underline{10} \rightarrow \underline{L}$$

$$\underline{\underline{[1 \times [F_1, F_2, F_3 \dots]]]}}$$

$$[\quad]$$

$$\underline{\underline{LL \quad ??}}$$

$$\underline{\underline{[F_1, F_2 \dots F_n]}}$$

$$\underline{\underline{1 \text{ row} \times 10 \text{ columns}}}$$